On The Gravitational Nature Of Time

Andrew Mueller[1*]

[1*]Independent Researcher, Milwaukee, 53219, WI, USA.

Corresponding author(s). E-mail(s): AndrewMueller414@gmail.com;

Abstract

We explore a modified set of coordinate transformations that describe time, cosmic inflation and gravitational acceleration as the same process as observed from different reference frames through an unwrapped Kaluza-Klein like geometry. I'LL HAVE AI DO THIS WHEN THE PAPER'S DONE.

Keywords: gravitation, time, relativity, inflation

1 Introduction

Albert Einstein was insistent on a static Universe during the period in which he wrote his specialized theory of relativity[1–4](SR). What is proposed here is largely a re-examination of the geometry of SR under the assumption that the $\mathbb{R}^3$ space between two arbitrary points elongates as time progresses.

This dilation is then bound to γ, giving a velocity dependent function for $\vec{g}$ and a magnitude that extends into a 4th spatial coordinate, y, similar in construction to an unwrapped Kaluza-Klein (KK) dimension[5, 6] with a known, non-zero derivative with respect to y. Through these coordinate transformations time, cosmic inflation and gravitational acceleration become the same process as observed from different reference frames in a space-time geometry where the 4th spatial KK dimension acts as a concurrent density axis.

This geometry gives an alternative model of concurrency and synchronicity as a classical notion of simultaneity is maintained by attributing the additional spatial separation to the 4th spatial dimension, $dy = v\tau\sqrt{\gamma^2 - 1}$ as shown in figure 7. While maintaining classical simultaneity resolves numerous temporal paradoxes without the need for selective reference frame preference, it produces a concurrent axis that describes the ultra-dense state of the early Universe as being separated only by some Δ along this 4th spatial axis.

1.1 The Lighthouse & The Clock-Tower

Let an event be defined classically, such that a single, instantaneous event represents a local measurement as $dt \to 0$.

Let there be two overlapping $\mathbb{R}^3$ coordinate systems, **S** and **U**. Let **S** and **U** represent an equi-temporal surface within a 4-dimensional hyperspace where $y = y^{(S)}$ and $y = y^{(U)}$ in each respective 3-dimensional surface.

This gives

$$f_{(x_i,y)} = \partial_y \mathcal{H} \tag{1}$$

where

$$f_{\left(x_i,y^{(A)}\right)} = \mathbf{A} \tag{2}$$

for an arbitrary **A** contained within $\mathcal{H}$.

Let **U** represent a 'future' state from the perspective of an observer in **S** so that

$$v^{(S)} = (t, x_\mu) \qquad v^{(U)} = (t + \Delta t, x_\mu) \tag{3}$$

where $\Delta t > 0$. Likewise, let **S** and **U** be bound such that

$$f_{\left(x_i,y^{(U)}\right)} = f_{\left(x_i,y^{(S)}\right)} + \int_{x_0^{(S)}}^{x_0^{(U)}} F_{(x)} \, \mathrm{d}t \tag{4}$$

where F is the sum of all forces acting on the point of space at x, integrated according to time, which in turn sums the change between the temporal surfaces **S** and **U**.

Let there also be two observers, A and B. Let A and B agree that B should travel at some *agreed upon* relativistic velocity v' between two arbitrary points in space contained within $\mathcal{H}$, p_1 and p_2. Let the experiment be arranged such that B passes p_1 in **S**, and p_2 in **U**.

Let A remain at relative rest with respect to p_1, and at p_2 place a time keeping device large enough to be visible from p_1.

We then make these assertions

1. To maintain a classical model of simultaneity A and B must agree upon the time at which B reaches p_2.
2. To satisfy existing experimental evidence for the additional γ factor, $\Delta t = \gamma \frac{d}{v}$.

If A and B are to disagree upon the time at which B reaches p_2 while requiring that $v = v'$, this single, instantaneous event (B reaching p_2) must occur more than once. For this reason we theorize that A and B *must* agree upon the time at which B reaches p_2 as a fundamental property of our reality.

As the additional factor of γ has been demonstrated experimentally on a number of occasions[7–11], we should conclude that both A and B observe B reaching p_2 in $\frac{d}{v}\gamma$. This gives a set of coordinate transformations such that for each relative observer

$$\Delta t = \overbrace{\frac{d \cdot \gamma}{v^{(U)}}}^{B} = \underbrace{\frac{d}{v^{(S)}} = \frac{d}{v^{(U)}\left(\frac{1}{\gamma}\right)}}_{A} \tag{5}$$

2 Theoretical Framework

We acknowledge two primary properties of time:

1. A driving rate of change, as fundamental as the $\mathbb{R}^3$ dimensions.
2. A 4th displacement coordinate.

Throughout the following derivations, we presume that time has no physical significance beyond the two properties described above. We presume that time acts only as a mathematical tool: $\Delta t = \frac{d}{v}$. In turn, we must attribute the 4th displacement coordinate to a position along the density axis that arises, y, and the driving rate of change to *motion*.

Likewise, we presume that in a co-moving frame of reference along the y axis, mechanical equations should limit at their Newtonian counterpart with only the relativistic portion of associated magnitudes protruding out of the containing temporal surface.

To remove time as a physically meaningful quantity, we let

$$\tau = \frac{\dot{x}}{\widehat{v}} \tag{6}$$

where $\dot{x}$ indicates not a derivative with respect to time, but rather a fundamental driving rate of change. Let $\widehat{v} = v \cdot 1t$, or the velocity in units of distance.

2.1 The Equivalence Principle

Here we examine these coordinate transformations in the context of the equivalence principle. While a well defined function remains elusive, a numerical intersection with the fine structure constant allows us to derive several important properties of an apparent electromagnetic function producing $\vec{g}$.

2.1.1 Earth's Kinematic Velocity From Electromagnetic Parameters

We define $v_{(\gamma)}$ to be

$$v_{(\gamma)} = c\sqrt{1 - \frac{1}{\gamma^2}} \tag{7}$$

And $\delta_{(M,R)}$ to be

$$\delta_{(M,R)} = \frac{d}{dR}g = 2\frac{GM}{R^3} \tag{8}$$

If $\delta_{(M,R)}$ is induced by spatial dilation, we should presume that this dilation is radial. We then find the magnitude of a single $\mathbb{R}^3$ axis by taking the inverse of the determinant of a diagonal 3×3 matrix:

$$\omega_{(M,R)} = \sqrt[3]{\frac{d}{dR} g} = \delta_{(M,R)}^{1/3} \tag{9}$$

Investigating this quantity for Earth gives

$$\omega_{\oplus} = 0.014537\ldots \tag{10}$$

While this approximates 2α, it more closely produces

$$\omega_{\oplus} = \alpha\,(2 - \alpha) \pm 0.0253\% \tag{11}$$

Then by expanding equation 11

$$\alpha\,(2 - \alpha) = \sqrt[3]{\frac{d}{dR} g} \tag{12}$$

Moving the electromagnetic parameters and R to one side before integrating with respect to R gives

$$R^4 \left(\frac{e^6 \left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10} \left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \right) = 1.272 \times 10^{21} \tag{13}$$

We then take the radius independent magnitude, isolating the additional electromagnetic parameters

$$\frac{e^6 \left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10} \left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{14}$$

As γ represents 1 plus some small ds, let us add 1 to these additional parameters and substitute γ_{em} into equation 7.

$$\gamma_{\text{em}} = 1 + \frac{e^6 \left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{4h^3} + \frac{3\mu_0 e^{10} \left(\frac{\mu_0}{\epsilon_0}\right)^{\frac{3}{2}}}{64\epsilon_0 h^5} - \frac{3\mu_0^2 e^8}{16\epsilon_0^2 h^4} - \frac{\mu_0^3 e^{12}}{256\epsilon_0^3 h^6} \tag{15}$$

$$v_{(\gamma_{\text{em}})} = 371,721 \text{ m/s} \tag{16}$$

This matches the 2018 Plank data[12] to within 0.514%, and is in rough agreement with general observation of the velocity necessary to induce the CMB dipole.

2.1.2 Solar ω

We expand equation 12 and define $\Gamma_{(\sigma)}$ to produce

$$\alpha = \Gamma_{(\sigma)} \left(\frac{d}{dR} g \right)^{1/3} \tag{17}$$

Where $\Gamma_{(\sigma)}$ is some unknown function of spin. While a well defined function remains elusive, we can take advantage of several characteristics for specific local bodies according to table 1 to derive certain characteristics of σ.

Table 1 Advantageous properties

Body	Trait	Value
Earth	The most mass dense.	5495 kg m^{-3}
Saturn	The least mass dense.	687 kg m^{-3}
Sun	The most massive.	1.988×10^{30} kg

In equation 12 we've demonstrated that for the most mass dense body in the system

$$\Gamma_{(\sigma),\oplus} \approx \Gamma_{(\alpha)} = \frac{1}{2 - \alpha} \tag{18}$$

We propose that $\Gamma_{(\sigma)}$ acts inside of the determinant. We then take the inverse of the determinant of a diagonal 3×3 matrix

$$\Gamma_{(\sigma),\odot} \propto \sqrt[3]{\Gamma_{(\sigma)}} \tag{19}$$

We then presume that σ is associated with spin. Due to the increased surface gravitational acceleration and the decreased rate of spin for the Sun, we approximate $\sigma = 0$.

$$\Gamma_{(\sigma),\odot} \approx \frac{1}{\sqrt[3]{2}} \tag{20}$$

Or

$$\alpha = \Gamma_{(\sigma),\odot} \sqrt[3]{\frac{d}{dR} g} = \sqrt[3]{\frac{1}{2} \frac{d}{dR} g_{\odot}} \pm 0.47\% \tag{21}$$

We then find through observation that for the least mass dense body in the system, $\Gamma_{(\sigma),\text{ħ}} \approx 1$.

This gives a set of 3 equations where the $\Gamma_{(\sigma)}$ term cancels for the dominant body in the system.

$$
\begin{aligned}
g_{\oplus} &= R_{\oplus}\left(\frac{\alpha^3\left(2-\alpha\right)^3}{2}\right) && \pm 0.008\% \\
g_{\odot} &= R_{\odot}\alpha^3 && \pm 1.426\% \\
g_{\hbar} &= \frac{1}{2}R_{\hbar}\alpha^3 && \pm 1.134\%
\end{aligned}
\tag{22}
$$

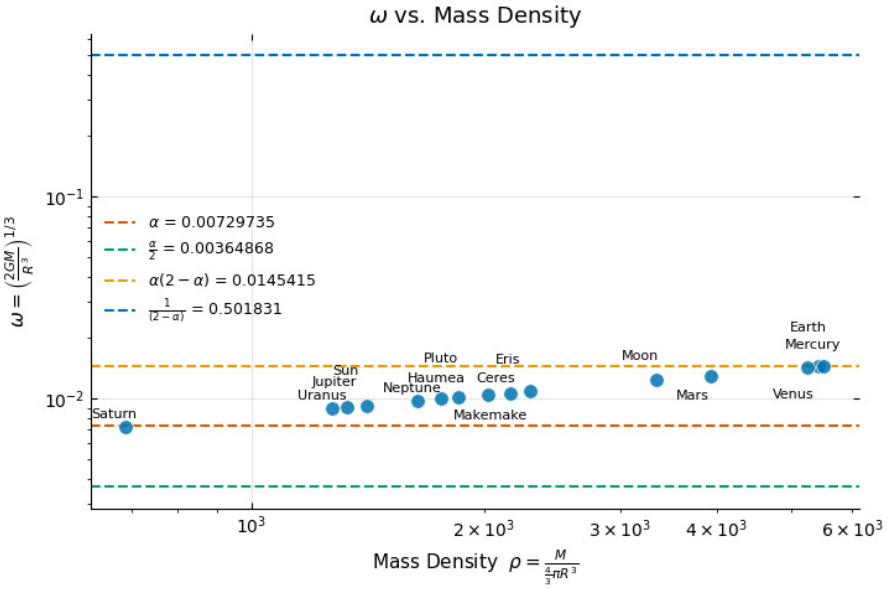


Fig. 1 The inverse of the determinant representing a change in volume, being equivalent to dy, plotted against mass density.

Which in turn produces a domain where $\alpha \lessapprox \omega \lessapprox \alpha\left(2-\alpha\right)$ as shown in figure 1.

2.2 Gravitational Parameters

Due to algebraic symmetries, it's trivial to derive our kinematic velocity through gravitational parameters. What is of interest however, is that we derive what we propose is a more fundamental velocity first. It is only after integrating motion in a manner that is analogous to time that we produce our kinematic velocity. We call this first velocity our *cosmological velocity*, and propose a new geometric interpretation of existing data that may resolve the SNe/CMB velocity discrepancy.

2.2.1 Cosmological Velocity

As γ is applied as a scalar, we take the average of the integral of $\delta_{(M,R)}$ to be

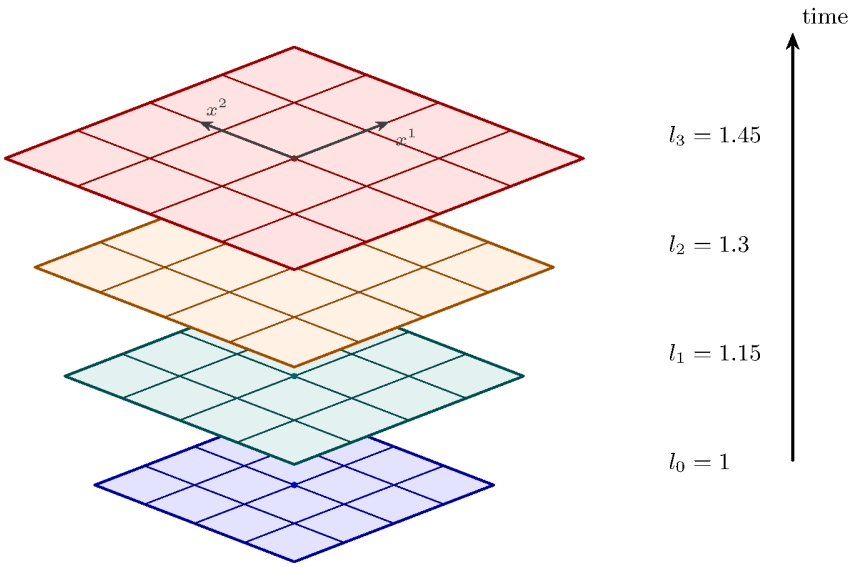


Fig. 2 A plane-based geometric simplification of the concurrent density axis found by these coordinate transformations.

$$\varphi_{(M,R)} = \frac{1}{R} \int \delta_{(M,R)} = 1 + G\frac{M}{R^3} \tag{23}$$

Substituting $\varphi_{(M,R)}$ into equation 7 gives

$$v_{\text{cosmo}} = c\sqrt{1 - \frac{1}{\varphi^2_{(M,R)}}} = 525,493.17 \text{ m/s} \tag{24}$$

While this value aligns closely with the correlation carried out by Gordon et al.[13], there remains large uncertainties inherent in SNe observation[14, 15].

As shown in figure 3, this function only holds for the upper bounds of mass density. We speculate that there may exist some yet unknown process in the formation of solar systems, particularly of our scale and location within the parent galaxy that leads to a natural function of mass density and velocity. While the details of this process have yet to be resolved, the coordinate transformations described here intrinsically link mass density to velocity as $\vec{g}$ becomes a function of v.

2.2.2 Kinematic Velocity

We apply the time-equivalent quantity in equation 6 to equation 23, making equation 23 time dependent. Integrating gives

$$\frac{1}{R}\int_0^R \int_0^x \delta\tau \; dR \; d\tau = 1 + \frac{1}{2}\frac{GM_\oplus}{R^3_\oplus}\left(\frac{\dot{x}}{\hat{x}}\right)^2 \tag{25}$$

We then substitute equation 25 into equation 7

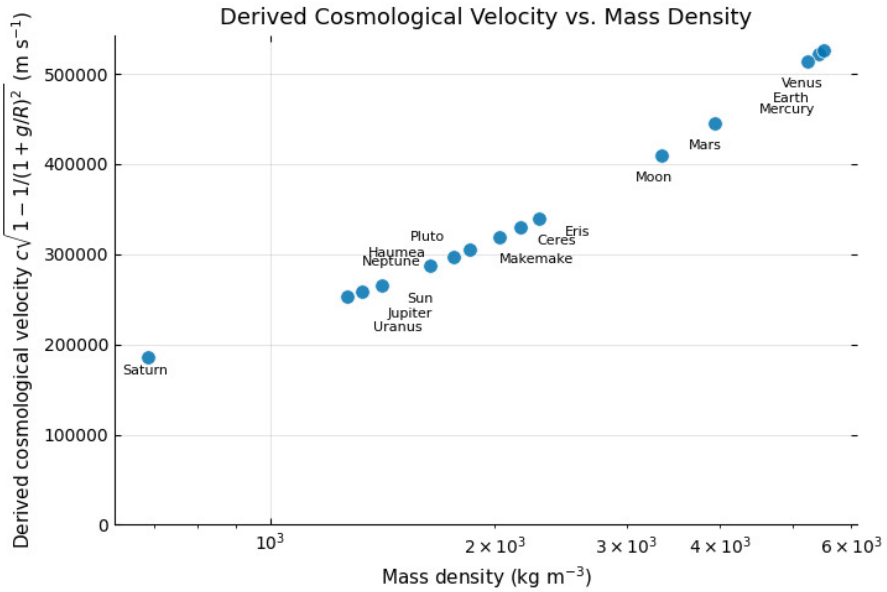


Fig. 3 A trigonometric derivation of a relativistic orbital velocity equation.

$$v_{\text{kinematic}} = c\sqrt{1 - \frac{1}{\left(1 + \frac{1}{2}G\frac{M}{R^3}\right)^2}} = 371,580 \text{ m/s} \tag{26}$$

which matches the 2018 Plank data to within 0.47%.

2.3 Orbital Velocity

2.3.1 Relativistic Stable Orbit Velocity

In the simplified single-body system considered here, the spatial dilation described above permits a trigonometric derivation of the stable-orbit velocity as an alternative to the conventional algebraic equality.

Figure 5 represents the geometric construction used in this derivation.

$$R_0^2 + v_{\text{orbit}}^2 d\tau^2 = \left(\int_0^{R_0} 2G\frac{M}{R^3}\right)^2 = (R_0 + g\ d\tau)^2 \tag{27}$$

Setting $\tau = 1$ gives

$$v_{(M,R),\text{orbit}} = \frac{\sqrt{GM\,(GM + 2R^3)}}{R^2} \tag{28}$$

For the comparison with the Milky Way, we adopt the enclosed Galactic baryonic mass reported by Cautun et al. [16]:

$$M_{\text{enc}} = M_{\text{bar}}(< R_0) = 4.44 \times 10^{10}\, M_\odot, \tag{29}$$

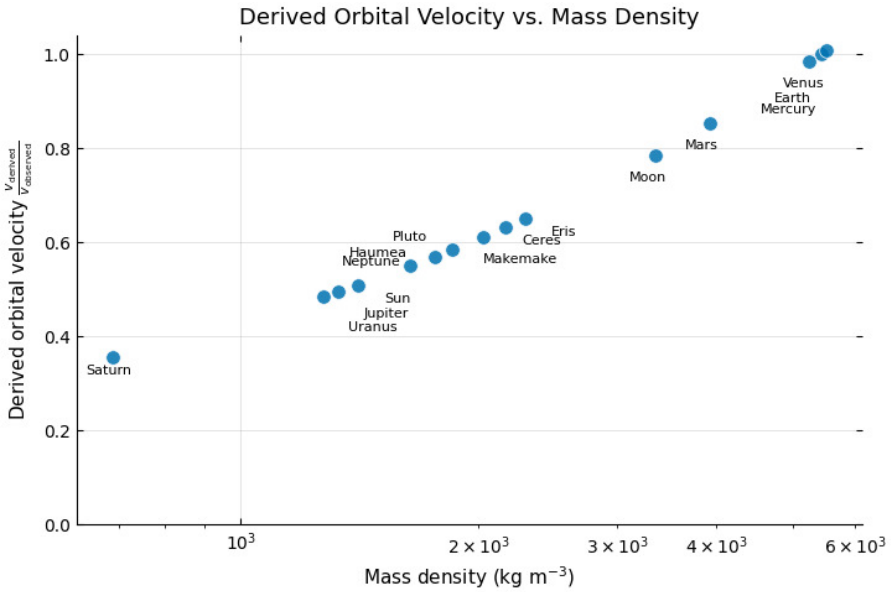


Fig. 4 A trigonometric derivation of a relativistic orbital velocity equation.

Here, $R_0 = 8.12\,\text{kpc}$ denotes the Sun's Galactocentric orbital radius[17, 18].

For comparison with observation, we use the solar-neighborhood circular velocity reported by Eilers et al. [19]. Using the adopted enclosed mass and solar orbital radius, the model gives the following orbital velocity:

$$v_{(M_{\text{enc}},R_{\text{orbit}}),\text{orbit}} = 216,874\ \text{m/s} = v_{\text{eilers}} \pm 5.59\% \tag{30}$$

The model velocity equals the comparison velocity at $M_{\text{enc}} \approx 4.95 \times 10^{10} M_{\odot}$.

The adopted mass scale is also consistent with the characteristic stellar mass scale of approximately $5 \times 10^{10}\,M_{\odot}$ identified by Yu et al. [20] in their FIRE-2 simulations of Milky Way-mass galaxies. Yu et al. associate this scale with a transition in galactic morphology from predominantly disc-dominated to increasingly spheroid-dominated systems. This result provides additional context for the adopted Galactic mass parameter.

2.3.2 Kinematic Velocity From Triginometric Orbit Equations

$$\mathrm{d}t = \frac{2R^3}{GM - 2R^3} = -1.000000768 \tag{31}$$

$$v_{(-\mathrm{d}t)} = 371,580\ \text{m/s} \tag{32}$$

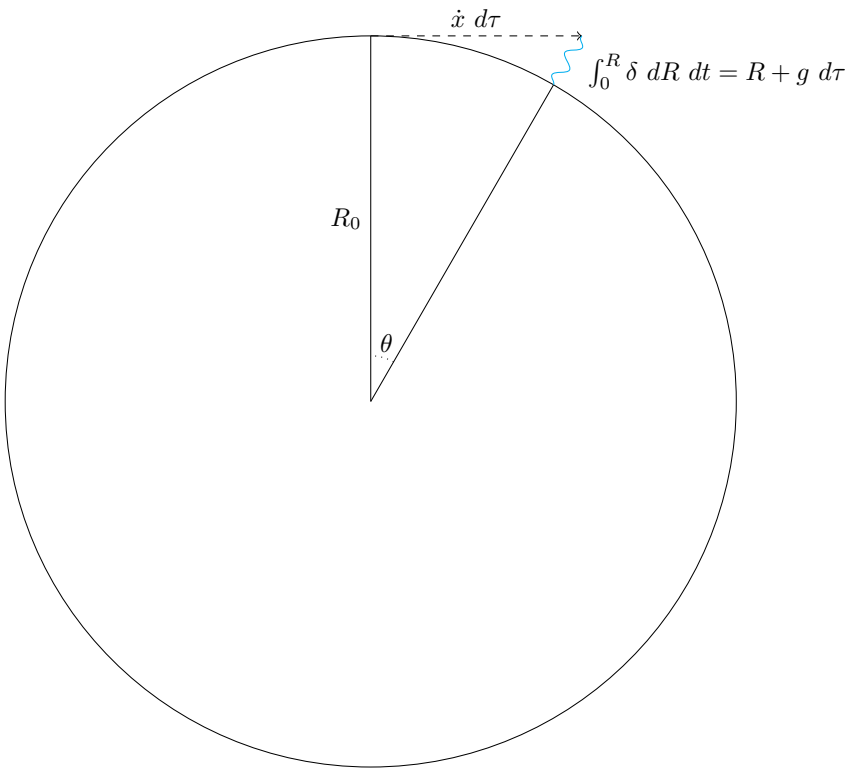


Fig. 5 A trigonometric derivation of a relativistic orbital velocity equation.

2.4 Fifth-Dimensional Spatial Displacement and Gravitational Force

2.4.1 Geometric Construction of the Fifth-Dimensional Displacement

We extend the ordinary three-dimensional spatial displacement into the additional spatial coordinate y. Let the magnitude of the ordinary spatial displacement be

$$dx = \sqrt{dx_i dx^i}, \qquad i = 1, 2, 3. \tag{33}$$

For a spatial velocity magnitude v, the displacement parameter τ defined in the theoretical framework gives

$$dx = v\tau. \tag{34}$$

The extended displacement forms the hypotenuse of a right triangle whose perpendicular components are dx and dy. Using the Lorentz factor γ already used in Equation 7, the hypotenuse is

$$dx^{(U)} = v\tau\gamma. \tag{35}$$

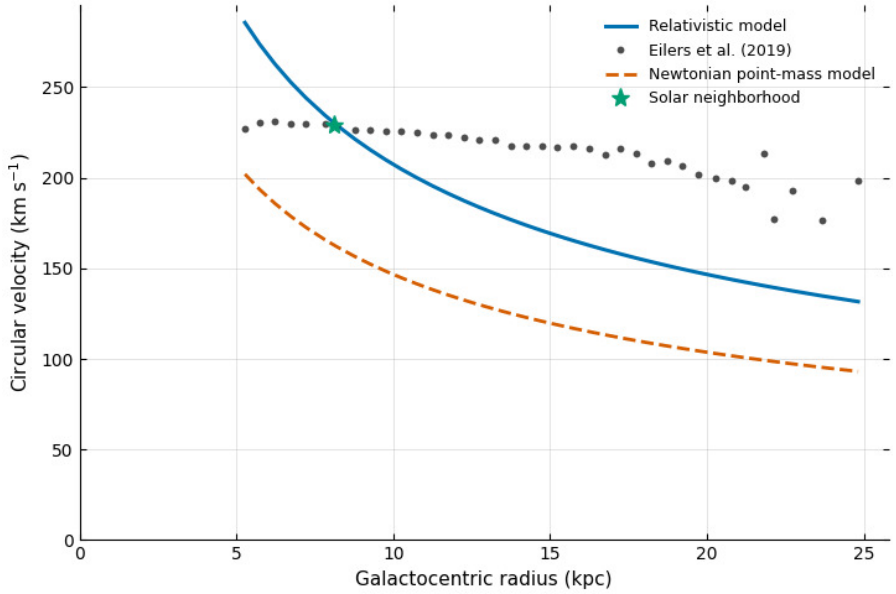


Fig. 6 Milky Way circular velocity as a function of galactocentric radius. Eilers et al. (2019) measurements are shown as points, alongside the relativistic and Newtonian point-mass model curves. The solar-neighborhood measurement is marked with a star; error bars indicate the reported uncertainties in galactocentric radius and circular velocity.

Applying the Pythagorean theorem,

$$\left(dx^{(U)}\right)^2 = dx^2 + dy^2, \tag{36}$$

and substituting the expressions for the ordinary and extended displacements gives

$$dy^2 = v^2\tau^2\gamma^2 - v^2\tau^2 \tag{37}$$

$$= v^2\tau^2(\gamma^2 - 1). \tag{38}$$

Taking the positive magnitude of the additional spatial displacement

$$dy = v\tau\sqrt{\gamma^2 - 1}. \tag{39}$$

Using the relation between v and γ in Equation 7, this becomes

$$\boxed{dy = \tau\frac{v^2}{\sqrt{c^2 - v^2}}} \tag{40}$$

Thus, the displacement along the additional spatial coordinate is expressed in terms of the ordinary spatial velocity magnitude and the displacement parameter.

Differentiating with respect to τ gives

$$\boxed{\frac{dy}{d\tau} = \frac{v^2}{\sqrt{c^2 - v^2}}} \tag{41}$$

2.5 Incorporation into the Kaluza–Klein Metric

We represent the additional spatial coordinate through the metric component G_{55}. For a diagonal metric, the five-dimensional line element is

$$dS^2 = g_{\mu\nu}dx^\mu dx^\nu + G_{55}dy^2. \tag{42}$$

The corresponding metric tensor has the block structure

$$\boxed{G_{AB} = \begin{pmatrix} g_{\mu\nu} & 0 \\ 0 & G_{55} \end{pmatrix}} \tag{43}$$

Here, G_{55} describes the metric scaling of the additional spatial coordinate. Its spatial variation determines how displacement along that coordinate couples to motion in the ordinary spatial coordinates.

For a radial coordinate R, the relevant Christoffel symbol is

$$\Gamma^R_{55} = -\frac{1}{2}g^{RR}\frac{\partial G_{55}}{\partial R}. \tag{44}$$

For a normalized radial coordinate with $g^{RR} = 1$,

$$\boxed{\Gamma^R_{55} = -\frac{1}{2}\frac{\partial G_{55}}{\partial R}} \tag{45}$$

2.6 Derivation of the Radial Force Contribution

The radial geodesic equation, using τ as its trajectory parameter and assuming no angular motion, is

$$\frac{d^2R}{d\tau^2} + \Gamma^R_{55}\left(\frac{dy}{d\tau}\right)^2 = 0. \tag{46}$$

Substituting Equation 45 yields

$$\frac{d^2R}{d\tau^2} = \frac{1}{2}\frac{\partial G_{55}}{\partial R}\left(\frac{dy}{d\tau}\right)^2. \tag{47}$$

Using the fifth-dimensional displacement rate in Equation 41,

$$\frac{d^2R}{d\tau^2} = \frac{1}{2}\frac{\partial G_{55}}{\partial R}\frac{v^4}{c^2 - v^2}. \tag{48}$$

Defining the corresponding generalized radial force contribution for a body of mass m as $\mathcal{F}_R = m\,d^2R/d\tau^2$, we obtain

$$\boxed{\mathcal{F}_R = \frac{mv^4}{2(c^2 - v^2)}\frac{\partial G_{55}}{\partial R}} \tag{49}$$

This relates the gradient of the fifth-dimensional metric component to its radial force contribution. The velocity-dependent factor follows from the geometric extension of the ordinary spatial displacement, while the metric gradient sets the strength and direction of the radial coupling.

Rearranging Equation 49 gives

$$\boxed{\frac{\partial G_{55}}{\partial R} = \frac{2(c^2 - v^2)}{mv^4}\mathcal{F}_R} \tag{50}$$

Thus, a known radial force constrains the spatial gradient of G_{55}, while a specified fifth-dimensional metric determines the corresponding radial force contribution through Equation 49.

The fifth-dimensional coordinate is treated as linear and non-compact; no periodic identification or compactification condition is imposed.

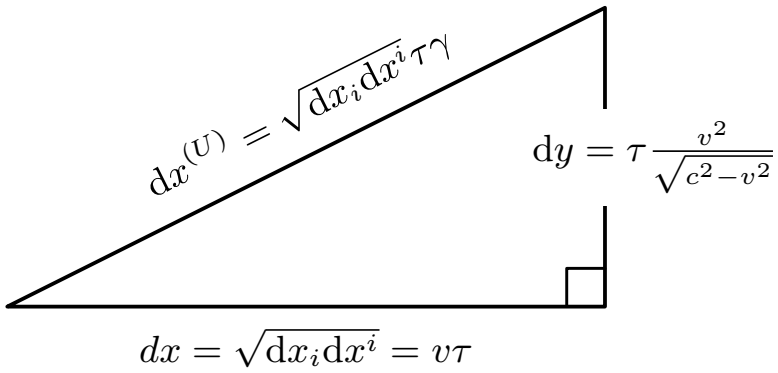


Fig. 7 Right-triangle construction used to represent the ordinary spatial displacement dx, the additional spatial displacement dy, and the extended displacement $dx^{(U)}$. The relation assigned to the hypotenuse is an assumption of the proposed geometric construction.

3 Discussion

3.1 The Appearance of Electromagnetic Parameters

We do not claim to derive a direct mechanical derivation for the fine structure constant, nor an explanation for it's appearance in section 2.1. We simply apply the coordinate transformations defined in section 2 and discover that α appears repeatedly. While the probability is difficult to quantize, we feel that this repeated appearance of α, appearing in gravitational approximations for all 3 of the local bodies with boundary like characteristics likely indicates a shared function which itself contains

α. Due to the hypersphere-like geometry that arises from this model, we propose further investigation into the boundary between Wyler's derivation[21] of α and $4+0$ space-time.

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